



POLITÉCNICA

Curso 2023-24
Seguridad de la Información
Exercises Block II
RSA and its properties



Objectives

Prepare for the exam:

- Topic 2: RSA, principles, encryption and decryption
- Topic 3: attacks

Exercises

Exercise 1.

Alice and Bob have the following RSA keys: $n_A = 91$, $e_A = 5$; $n_B = 85$, $e_B = 7$.

- Find Alice's private key d_A
- Find Bob's private key d_B
- Encrypt the message $M = 24$ from Alice to Bob
- Decrypt the message that Bob receives.

Exercise 2.

In a textbook example, a Client and a Server interchange a 6-bit session key using RSA with a 9-bit modulus. The key is generated by the Client and sent to the Server.

The data of the two parties are

- Client (C): $p = 13$; $q = 31$; $e = 13$; $d = 277$
- Server (S): $p = 19$; $q = 23$; $e = 5$; $d = 317$

- The session key is $K_S = 61$. Calculate its encryption, and describe the decrypting operation at the receiver's end.
- Explain what happens if the RSA keys of Client and Server above are swapped? Is the key exchange better or worse than before?